

PFM 203 Course Outline

PFM 203 — MTEF Modelling

Course Outline

School	02 — Macroeconomics and Public Finance
Level	200 (analyst / practitioner)
Hours	30 (indicative band 24–32)
Shape	5 modules · 10 sessions · 6 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101. MAC 202 FINEX Macro-Fiscal Modelling, or equivalent familiarity with where a macro-fiscal envelope comes from, is strongly recommended. Julia at the level of running a script and reading a configuration file is required; no Julia programming is assumed.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-24 · 12-month cycle

What this course is

PFM 203 teaches the AICTEM MTEF engine: a 6,552-line Julia implementation across 18 layers that reads a reconciled approved budget from the Budget module, takes a macro-fiscal envelope from FINEX, projects what current policy costs on drivers rather than growth rates, assesses the maintenance need from the asset base, carries the commitment register, cuts sector ceilings, selects a multi-year investment portfolio by mixed-integer programme, and feeds the result back to FINEX until the loop converges.

It is built on the implemented engine and its published run, not on a description of one. Every identity taught is quoted from its source file; every number is read from a published table; every limitation is named where a learner will meet it.

The distinction the whole course turns on. PRESERVING current services is not the same as EXPANDING them, and a forward estimate that cannot tell them apart cannot support a policy discussion. If the wage bill, the maintenance need and the committed project schedule are not each estimated separately, then every number after them is a single opaque total and the only available conversation is about that total's growth rate.

Who it is for

Budget directorate, planning and policy analysts in ministries of finance; sector planning and budgeting officers who receive a ceiling and need to know where it came from; public investment management staff; and EDIS developers maintaining the MTEF engine or its sector gateway.

What this course does not claim

This course does not teach budget preparation. The approved budget is an input; PFM 204 teaches the module that produces it.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	What an MTEF is, and what this engine is; The two anchors: budget and FINEX	01–02	01	5.4
M2	The accounting framework;	03–04	01–02	5.4
M3	Forward estimates			
M3	Maintenance and the asset base;	05–06	03–04	5.4
	Commitments, projects and the pipeline			
M4	Ceilings, allocation and the portfolio; Sector integration and the bridges	07–08	04–05	7.4
M5	Scenarios, convergence and diagnostics;	09–10	06	5.4
	Interpretation and reporting			
ASM	Assessment and revision	—	—	1.0
	Total	10	6	30.0

The 10 sessions and 6 laboratories, in teaching order, and the subject matter of each:

M1 What an MTEF is, and what this engine is; The two anchors: budget and FINEX — Sessions 01–02 · 5.4 h · Laboratory 01

- **01 What an MTEF is, and what this engine is** — the medium-term expenditure framework as a fiscal instrument; scope, non-scope and the boundary rule; the pipeline, and the loop; the eight rules the model will not break

- *Lab 01 Reconcile a base year* — run the engine, read the FY0 reconciliation, and prove the base year is the approved budget rather than an approximation of it
- **02 The two anchors: Budget and FINEX** — the FY0 baseline: reconciling to the approved budget; two fiscal definitions, reconciled rather than merged; the expenditure envelope; what the envelope is not

M2 The accounting framework; Forward estimates — Sessions 03–04 · 5.4 h · Laboratory 01–02

- **03 The accounting framework** — the requirement identity; recurrent, development and the economic columns; the funding gap; the identities, and the checks that close them
- **04 Forward estimates** — driver-based forward estimates; the wage bill; non-wage: service volumes and unit costs; reading a forward estimate
- *Lab 02 Build a forward estimate* — read the driver configuration, identify which blocks used a driver and which fell back, and reproduce one sector’s forward path by hand

M3 Maintenance and the asset base; Commitments, projects and the pipeline — Sessions 05–06 · 5.4 h · Laboratory 03–04

- **05 Maintenance and the asset base** — assessed maintenance need; the under-maintenance penalty; rehabilitation, and the Utopia maintenance finding; maintenance floors and the maintenance-first scenario
- *Lab 03 Assess a maintenance backlog* — compute the assessed maintenance need, measure the under-maintenance penalty, and test how far the answer moves when the capital-stock assumption changes
- **06 Commitments, projects and the pipeline** — ongoing project commitments; counterpart funding and external financing; lifecycle cost and future OPEX; the pipeline and the token-provision problem, and more
- *Lab 04 Run the portfolio optimiser* — select a portfolio, read the binding constraints, and make the problem infeasible on purpose so that the failure mode is familiar

M4 Ceilings, allocation and the portfolio; Sector integration and the bridges — Sessions 07–08 · 7.4 h · Laboratory 04–05

- **07 Ceilings, allocation and the portfolio** — preliminary and final ceilings; protection, floors and overrides; the portfolio optimisation problem; binding constraints and infeasibility, and more
- **08 Sector integration and the bridges** — the standard sector interface; primary ownership, and why it prevents double counting; cross-sector dependencies; the CDCGE handover, and more
- *Lab 05 Construct a sector submission* — build a sector fiscal submission by hand against the gateway contract, and see the requirement change from budget-derived to costed

M5 Scenarios, convergence and diagnostics; Interpretation and reporting — Sessions 09–10 · 5.4 h · Laboratory 06

- **09 Scenarios, convergence and diagnostics** — what a scenario may and may not do; the sixteen scenarios; FINEX feedback and convergence; diagnostics: failures against findings, and more
- *Lab 06 Run and compare scenarios* — run the sweep, compare scenarios honestly, attempt a forbidden scenario, and write a contract-compliant brief
- **10 Interpretation and reporting** — the reporting contract; common errors; A worked ministerial brief

What a graduate can do

1. Place the MTEF engine against the Budget module and FINEX using the boundary rule, and say for any question which module owns it.
2. Reconcile a base year exactly at three levels and explain why the tolerance is floating-point noise.
3. Put two fiscal definitions on one basis, compute the coverage ratio and say what a moving ratio would mean.
4. Write the requirement identity, define its seven terms, and map each to the budget columns with its double-count risk.
5. Build a driver-based forward estimate, identify the fallbacks used, and compare requirement growth with nominal GDP growth.
6. Assess a maintenance backlog, compute the under-maintenance penalty, and state what the finding does and does not establish.
7. Compute a lifecycle fiscal cost and apply the commissioning rule.
8. Construct preliminary and final ceilings, compute the discretionary share and explain the protection hierarchy.
9. Read a portfolio result, report what binds, and diagnose an infeasible run without relaxing a constraint.
10. Use the sector gateway correctly, distinguishing a submission from an indicator link, and explain primary ownership.

The 14 formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	83 items across 10 modules
Laboratory exercises	30 %	6 laboratories, equally weighted, marked on method, interpretation, honesty about limits and communication
Applied assignments	25 %	base-year reconciliation note; maintenance backlog assessment; portfolio and constraints; scenario brief
Final assessment	30 %	Three hours, open book: definitions and identities; calculation; interpretation; reporting

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been recorded with its actual value: an unchecked result is no result.

Materials and set-up

A 196-page technical manual; 10 lecture decks and 6 laboratory decks; a laboratory manual, instructor guide and learner workbook; a 83-item quiz bank, 4 assignments and a final assessment with its marking

guide; 5 datasets; 5 code exercises; 4 case studies; and the lms package. The README lists every file. Every figure in this course is traceable to a published source or a committed model output; the sources are registered in `COURSE_VERSION.json` and the limitations in `COURSE_BUILD_REPORT.md`. Laboratories need only the extracts shipped in `08_Data/`.

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